Data Dictionary — Weather (NASA POWER)

Master Thesis: Crop Harvesting Prediction · Winter wheat harvest-window forecasting · France

Source: NASA POWER (Prediction Of Worldwide Energy Resources) · **API:** Daily Point, community AG · **Region:** Centre-Val de Loire (6 departments)
Coverage: 1 Oct 2019 → 31 Jul 2024 · **Rows:** 10,596 (6 points × 1,766 days) · **Columns:** 30

1. Overview

This dataset provides daily meteorological data for the Centre-Val de Loire region, retrieved from NASA POWER, plus a set of agronomic features derived for winter-wheat harvest prediction. Because weather varies little between nearby parcels, the region is represented by **6 points — one per department** (the préfecture city of each), rather than one point per parcel. Each parcel will later be linked to the weather of its department.

> **Why 6 points and not one per parcel?** Meteorology is a large-scale phenomenon: two parcels 10–20 km apart experience almost identical temperature and rainfall. Sampling one point per department captures the region's north–south gradient (oceanic in the north, more continental in the south) while avoiding redundant data — unlike vegetation (NDVI), which genuinely differs parcel by parcel.

The 6 weather points (one per department)

Dept	City (point)	Lat	Lon	Position
28	Chartres	48.444	1.489	North (Eure-et-Loir)
45	Orléans	47.902	1.909	North-East (Loiret)
41	Blois	47.586	1.336	Centre (Loir-et-Cher)
37	Tours	47.394	0.689	West (Indre-et-Loire)
18	Bourges	47.081	2.399	East (Cher)
36	Châteauroux	46.811	1.690	South (Indre)

2. Column dictionary

KEY *identifier / time* **RAW** *raw NASA POWER variable* **DERIVED** *engineered feature*

Identifiers & time

Column	Type	Unit	Description
dept KEY	text	—	Department code (18, 28, 36, 37, 41, 45). Links each weather point to a parcel's department.
ville KEY	text	—	Reference city (department préfecture) used as the weather point.
lat KEY	float	degrees	Latitude of the point (WGS84).
lon KEY	float	degrees	Longitude of the point (WGS84).
date KEY	date	YYYY-MM-DD	Observation day.
saison KEY	int	year	Agricultural campaign, identified by harvest year. Runs Oct (N-1) → Jul (N). E.g. 1 Oct 2019 → saison 2020.
doy KEY	int	1–366	Day of year — position in the calendar year.

Raw variables (NASA POWER)

Column	Type	Unit	Description

T2M_MAX RAW	float	°C	Daily maximum air temperature at 2 m. Drives GDD and heat-day counts.
T2M_MIN RAW	float	°C	Daily minimum air temperature at 2 m. Drives GDD; frost indicator.
T2M RAW	float	°C	Daily mean air temperature at 2 m.
T2MDEW RAW	float	°C	Dew point temperature at 2 m. Humidity / vapour-pressure indicator.
PRECTOTCORR RAW	float	mm/day	Bias-corrected total precipitation. Core water-input variable.
RH2M RAW	float	%	Relative humidity at 2 m. Disease pressure, grain drying.
ALLSKY_SFC_SW_DWN RAW	float	MJ/m²/day	All-sky surface shortwave downward irradiance (solar radiation). Photosynthesis; input to ET0.
WS2M RAW	float	m/s	Wind speed at 2 m. Input to ET0; lodging risk.

Derived features — thermal

Column	Type	Unit	Description
GDD DERIVED	float	°C·day	Daily Growing Degree Days, base 0 °C: max(0, (Tmax+Tmin)/2 − 0).
GDD_cumul DERIVED	float	°C·day	Cumulative GDD (base 0) since the start of the campaign, per department. **Key development driver.**
GDD5 DERIVED	float	°C·day	Daily GDD, base 5 °C (standard wheat variant): max(0, (Tmax+Tmin)/2 − 5).
GDD5_cumul DERIVED	float	°C·day	Cumulative GDD (base 5) per campaign and department.
amplitude_therm DERIVED	float	°C	Daily thermal amplitude (Tmax − Tmin). Stress and grain-quality indicator.
jour_chaud DERIVED	int	0/1	Hot-day flag: 1 if Tmax > 25 °C, else 0.
jours_chauds_cumul DERIVED	int	days	Cumulative count of hot days per campaign. High counts accelerate ripening (grain scorching).

Derived features — water balance

Column	Type	Unit	Description
ET0 DERIVED	float	mm/day	Reference evapotranspiration, **Penman-Monteith FAO-56**. Evaporative demand.
ET0_cumul DERIVED	float	mm	Cumulative ET0 per campaign and department.
precip DERIVED	float	mm/day	Copy of PRECTOTCORR kept for clarity in derived block.
precip_cumul DERIVED	float	mm	Cumulative precipitation per campaign and department.
bilan_hydrique DERIVED	float	mm/day	Daily water balance: precipitation − ET0. Negative = water deficit.
bilan_hydrique_cumul DERIVED	float	mm	Cumulative water balance per campaign. **Real water-stress indicator.**
pluie_30j DERIVED	float	mm	Rolling 30-day precipitation sum. Recent moisture (grain drying before harvest).

Derived features — energy

Column	Type	Unit	Description
rayonnement_cumul DERIVED	float	MJ/m²	Cumulative solar radiation per campaign and department. Total photosynthetic energy received.

3. Methodology notes

- **Agricultural campaign:** winter wheat is sown in Oct (year N-1) and harvested in July (year N). The "saison" field labels each campaign by its harvest year, so all cumulative features reset at the start of each campaign.

- **GDD (Growing Degree Days):** measure accumulated heat, the true driver of wheat development (the crop progresses with heat, not with the calendar). Both base 0 °C and base 5 °C are provided; base 0 is standard for French winter wheat (McMaster & Wilhelm, 1997).
- **ET0 (Penman-Monteith FAO-56):** the FAO reference method, computed from temperature, wind, solar radiation and humidity. Validated against expected ranges: ~0.2–1 mm/day in winter, ~4–6 mm/day in July.
- **Solar radiation unit:** NASA POWER delivers ALLSKY_SFC_SW_DWN directly in MJ/m^2/day (no kWh→MJ conversion needed) — an early version of the ET0 code mistakenly applied a ×3.6 factor and was corrected.
- **Cumulative features** are computed per (department × campaign) so they correctly reset each season and never mix departments.

4. Data quality

- **No missing values** in any column (10,596 / 10,596 rows complete).
- **No NASA POWER fill codes (−999)** detected in any raw variable.
- **Physical plausibility verified:** ET0 seasonal pattern correct; temperatures, rainfall and radiation within realistic ranges for the region.

5. Project usage

- **Join key:** each wheat parcel is linked to weather by its *department* and *date* (or aggregated 10/15-day window).
- **Most predictive features expected:** GDD_cumul / GDD5_cumul (development stage), bilan_hydrique_cumul (water stress), jours_chauds_cumul (ripening acceleration), pluie_30j (pre-harvest moisture).
- **Worked signal:** in 2023, Chartres reached GDD5_cumul ≈ 2,000 with a strongly negative cumulative water balance and falling 30-day rainfall by early June — conditions consistent with the observed early harvest (3 July 2023).